TCGA-BRCA: Gene Expression and PAM50 Visualization

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Introduction

Breast cancer represents a major global public health challenge requiring multinational collaboration in scientific research and pharmaceutical development. In 2024, approximately 2.4 million women were diagnosed with breast cancer worldwide, with an estimated 694,000 associated deaths (World Health Organization, 2026). Despite advances in screening and treatment, breast cancer remains a biologically heterogeneous disease, exhibiting considerable variation in molecular characteristics and clinical behavior.

Through the contributions of scientists worldwide, gene-expression research has identified distinct molecular subtypes, including Luminal A, Luminal B, HER2-enriched, and Basal-like breast cancers (Parker et al., 2009). These classifications demonstrate differences in tumor biology and provide researchers with a framework for investigating variation in gene expression. The Cancer Genome Atlas (TCGA) has established an extensive molecular resource for investigating breast cancer heterogeneity (Cancer Genome Atlas Network, 2012).

Two genes, ESR1 and ERBB2, were selected for their associations with estrogen signaling and HER2-related growth pathways, respectively (Cancer Genome Atlas Network, 2012). ESR1 (Estrogen Receptor 1) encodes estrogen receptor alpha, which regulates hormone-dependent gene expression and is strongly associated with luminal breast cancer subtypes (Parker et al., 2009). ERBB2 (HER2) encodes a receptor tyrosine kinase involved in cellular growth and survival, with amplification or overexpression associated with HER2-positive breast cancer (Cancer Genome Atlas Network, 2012).

Methods

We examined the supplied gene-expression matrix (`counts.csv`) and associated sample annotations (`meta_data.csv`). The results presented here are based, in whole or in part, on data

generated by the TCGA Research Network.

TCGA was made possible through the voluntary contributions of patients who consented to the use of their biological specimens and associated clinical information for cancer research. These donated tissue samples and accompanying clinical data have enabled significant advances in the genomic and molecular characterization of cancer. TCGA established informed-consent procedures and data-access safeguards to protect participants' rights while supporting scientific research. These patient contributions continue to advance our understanding of breast cancer heterogeneity.

[State the verified data origin and assay, how sample identifiers were matched, the selected sample types, exclusions, and the exact subtype field.]

[Describe any gene filtering, normalization or transformation, Z-score calculations, and how genes were selected for visualization.]

Descriptive statistics and a heatmap were used to characterize the selected cohort. An exploratory volcano plot was generated to illustrate differences between specified sample groups.

[State the actual comparison, calculation of effect sizes and p-values, and any multiple-testing adjustment.]

Unless an appropriate count-based statistical model was fitted, these comparisons should be interpreted as exploratory rather than confirmatory (Benjamini & Hochberg, 1995; Love et al., 2014).

Results

[Report the final matched sample count and molecular subtype distribution, including denominators and missing annotations.]

[Report reproducible descriptive findings from the gene-expression summaries.]

[Reference the final heatmap and briefly describe its most relevant expression patterns, noting that Z-score tails do not represent statistical significance.]

[Reference the final volcano plot and identify the comparison groups, axes, thresholds,

and methodological limitations. Do not interpret exploratory highlights as validated differential-expression findings.]

Figures

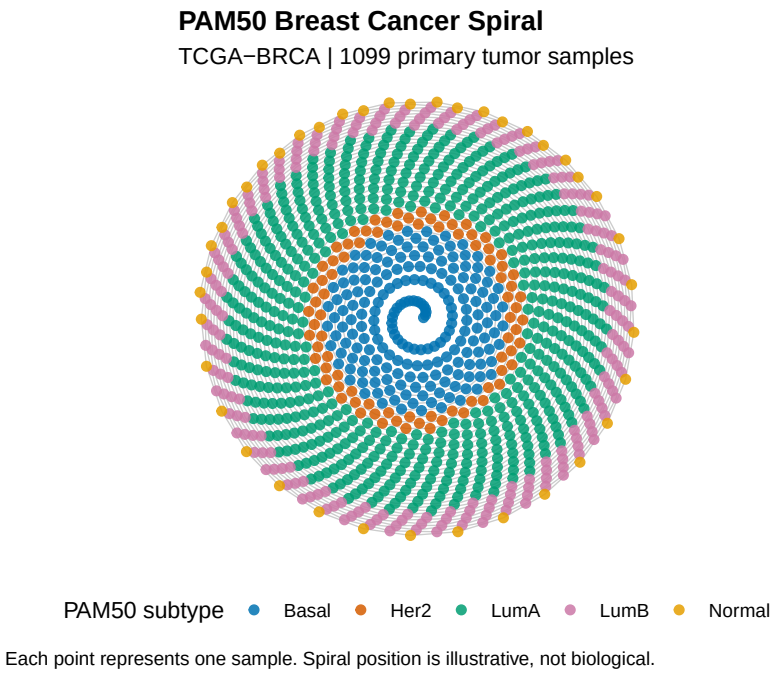


Figure 1: Visualization of PAM50 molecular subtypes in the TCGA-BRCA cohort.

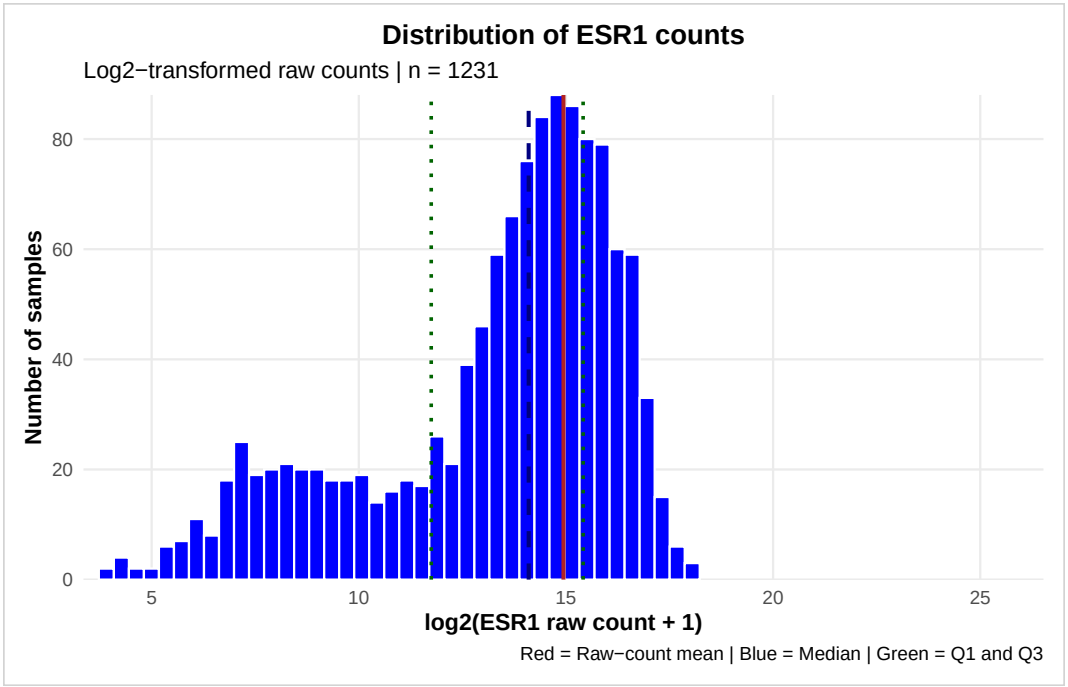


Figure 2: Distribution of ESR1 gene-expression values.

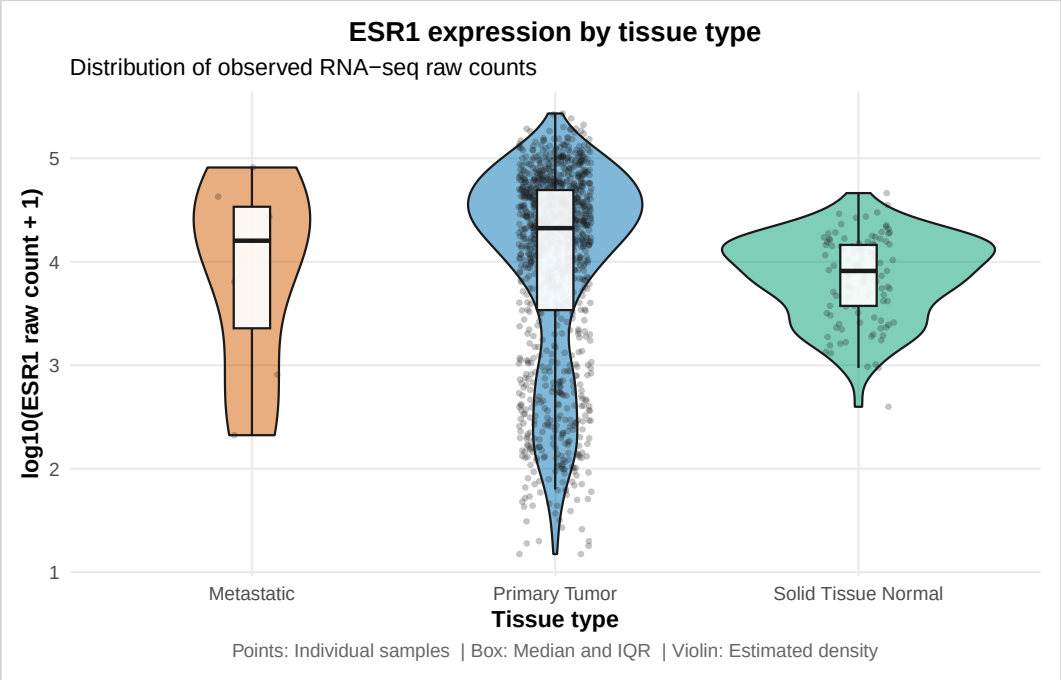


Figure 3: ESR1 expression across sample types.

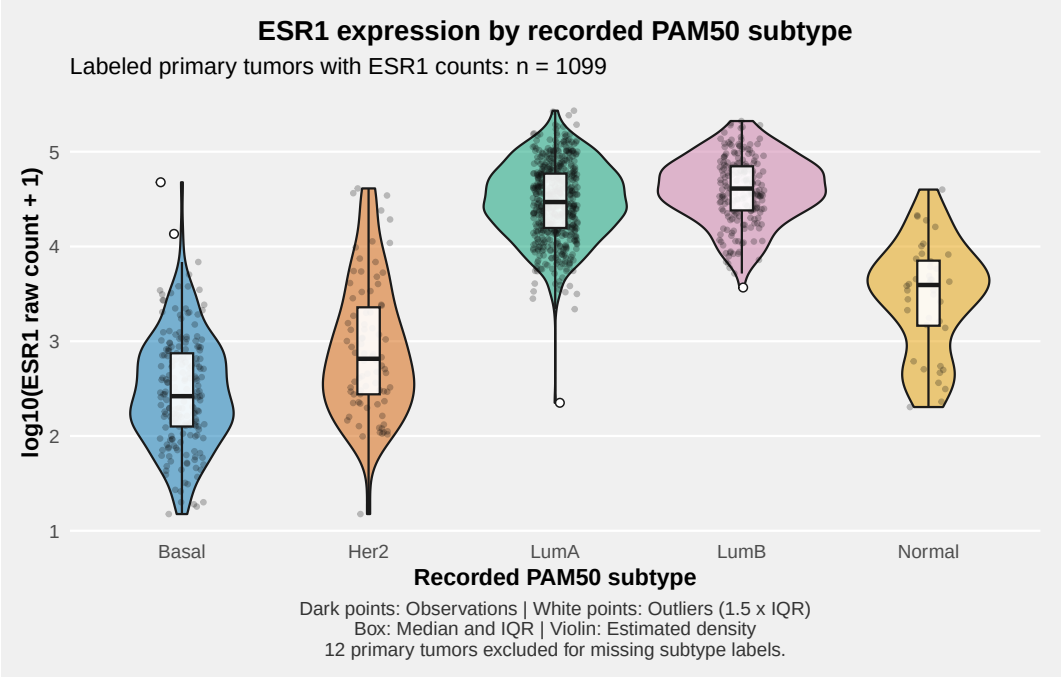


Figure 4: ESR1 expression across PAM50 molecular subtypes.

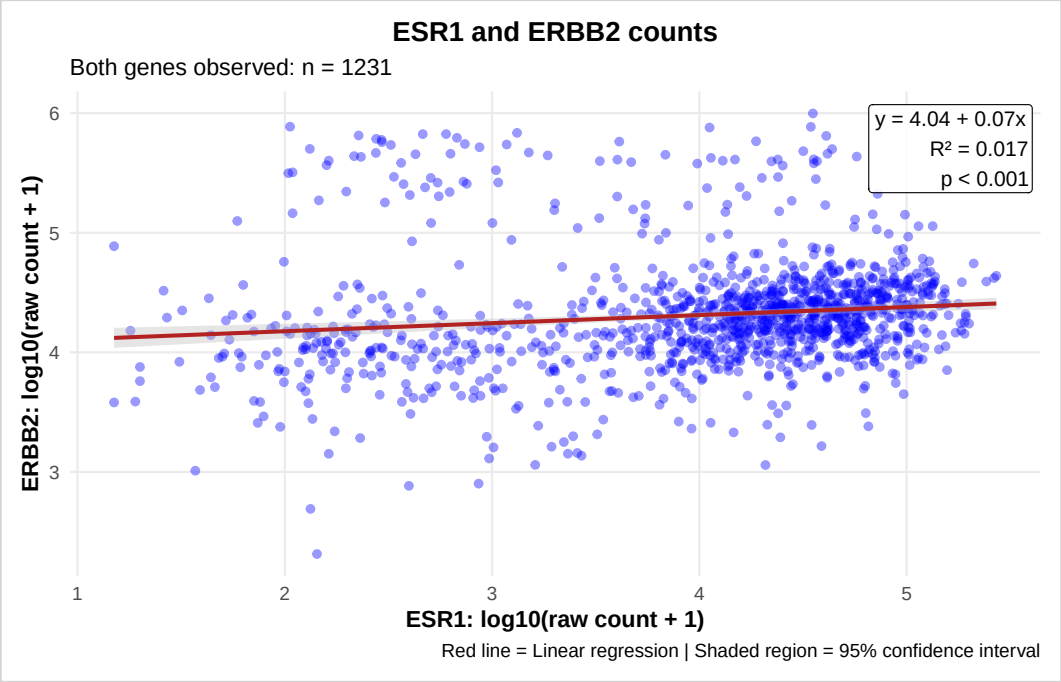


Figure 5: Exploratory gene-expression scatterplot.

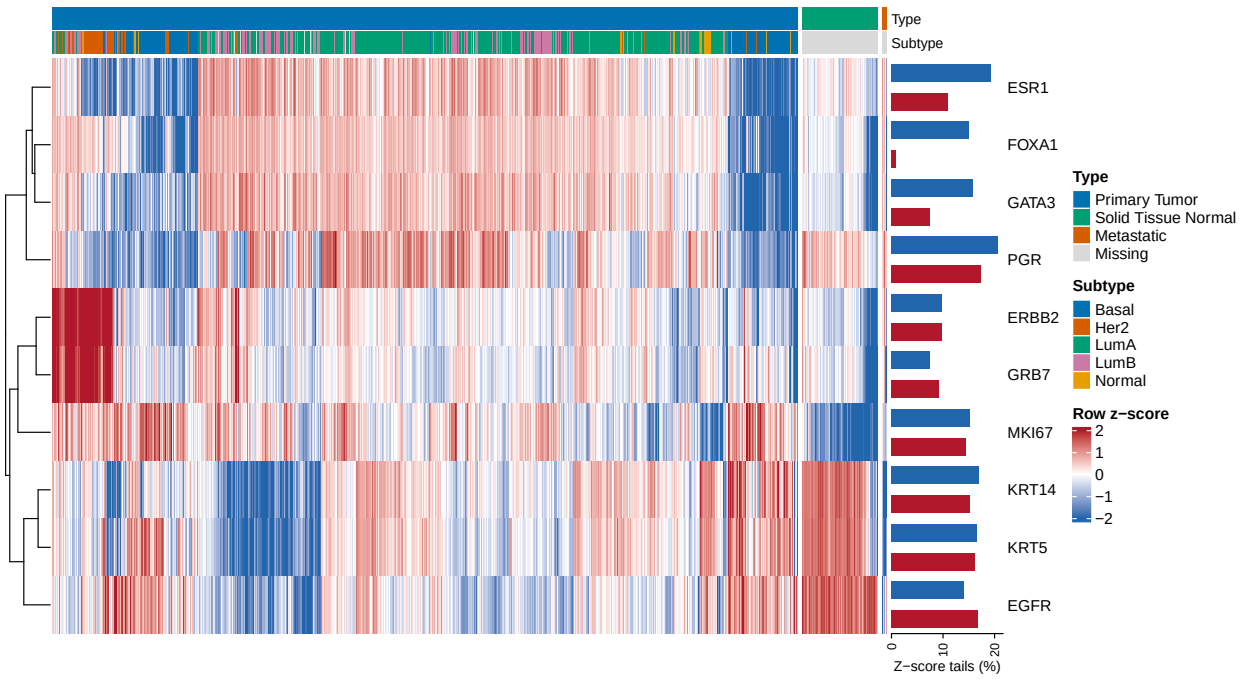


Figure 6: Heatmap of selected breast cancer-associated gene-expression patterns.

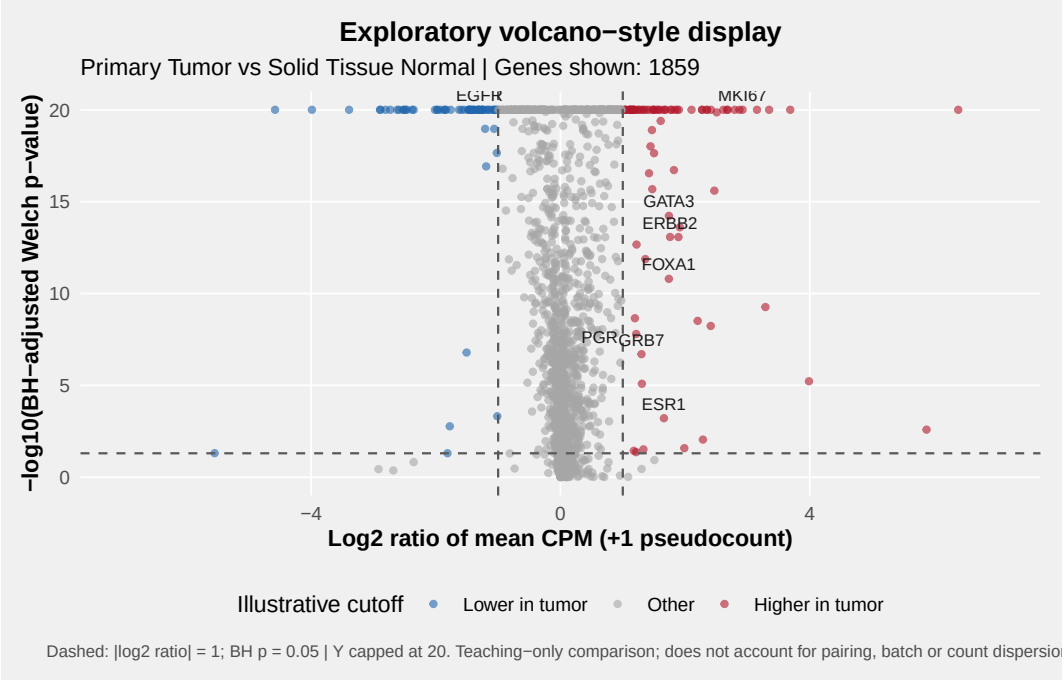


Figure 7: Exploratory volcano plot of gene-expression differences between comparison groups.

Tables

Table 1: Primary tumors with a recorded PAM50 subtype ($n = 1{,}099$).

Subtype	Samples	Percent (%)
Basal	197	17.9
Her2	82	7.5
LumA	571	52.0
LumB	209	19.0
Normal	40	3.6

Table 2: ESR1 and ERBB2 raw-count descriptive statistics.

Gene	N	Missing	Mean	Median	SD	IQR	Min	Max
ESR1	1,231	0	31,442.2	17,561	38,261.2	40,242.5	14	270,977
ERBB2	1,231	0	46,982.5	19,125	105,287.5	19,287.0	205	995,067

Table 3: ESR1 raw-count statistics by PAM50 molecular subtype.

Subtype	N	Mean	Median	SD	IQR	Min	Max
Basal	197	924.5	262.0	3,589.1	620.0	14	47,589
Her2	82	3,414.2	652.5	7,745.6	2,010.8	14	40,971
LumA	571	43,811.7	29,436.0	39,799.2	43,034.5	223	270,977
LumB	209	53,197.4	40,849.0	41,119.8	46,128.0	3,686	210,656
Normal	40	6,066.0	3,922.5	7,775.7	5,626.5	201	39,845

AI Disclosure

AI was used to type in the file names for the figures. The main file before modifications was cleaned by AI from an old manuscript to empty the file, and pre-cook with reporting basic instructions - NO AI USED FOR MANUAL TYPE UP. After template was cooked, this was hand written, and typed with the exception of the file names. VIEW GITHUB for documentation chain.

References

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Cancer Genome Atlas Network. (2012). Comprehensive molecular portraits of human breast tumours. *Nature*, *490*(7418), 61–70. https://doi.org/10.1038/nature11412

Love, M. I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for rna-seq data with DESeq2. *Genome Biology*, *15*(12), 550. https://doi.org/10.1186/s13059-014-0550-8

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